

4.2 上下文无关文法

$$\begin{aligned}
 1. \quad & S \rightarrow (L) | a \\
 & L \rightarrow L, S | S \\
 & S \rightarrow (L) | a \\
 & L \rightarrow SL' \\
 & L' \rightarrow ,SL' | \epsilon
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & S \rightarrow AbB | A \\
 & A \rightarrow AB | caB | B \\
 & B \rightarrow Aa | b
 \end{aligned}$$

① 将B的定义代入A-产生式, 得

$$A \rightarrow \underbrace{AAa}_{\alpha_1} | \underbrace{Ab}_{\alpha_2} | \underbrace{caAa}_{\beta_1} | \underbrace{cab}_{\beta_2} | \underbrace{Aa}_{\alpha_3} | \underbrace{b}_{\beta_3}$$

② 消除A产生式而直接左递归

$$\begin{aligned}
 A &\rightarrow \underbrace{caAa}_{\beta_1} A' | \underbrace{cab}_{\beta_2} A' | \underbrace{bA'}_{\beta_3} \\
 A' &\rightarrow \underbrace{Aa}_{\alpha_1} A' | \underbrace{bA'}_{\alpha_2} | \underbrace{aA'}_{\alpha_3} | \epsilon
 \end{aligned}$$

③ 消除左公因子, 最后写为:

$$\begin{aligned}
 S &\rightarrow A S' \\
 S' &\rightarrow bB | \epsilon \\
 A &\rightarrow caBA' | bA' \\
 A' &\rightarrow BA' | aA' | \epsilon \\
 B &\rightarrow Aa | b
 \end{aligned}$$

$$\begin{aligned}
 3. \quad & S \rightarrow aFbM | F \\
 & F \rightarrow M | abc \\
 & M \rightarrow abF | c
 \end{aligned}$$

① 将M的定义代入F、S的产生式

$$\begin{aligned}
 & F \rightarrow abF | c | abc \\
 \Rightarrow & F \rightarrow abF | c \\
 & S \rightarrow aFbabF | aFbc | F \\
 \Rightarrow & S \rightarrow aFbF | F
 \end{aligned}$$

② 消除形如 $A \rightarrow BA$ 的产生式

$$\begin{aligned}
 S &\rightarrow aFbF | abF | c \\
 F &\rightarrow abF | c
 \end{aligned}$$

③ 提取左公因子

$$\begin{aligned}
 S &\rightarrow aS' | c \\
 S' &\rightarrow FbF | bF \\
 F &\rightarrow abF | c
 \end{aligned}$$